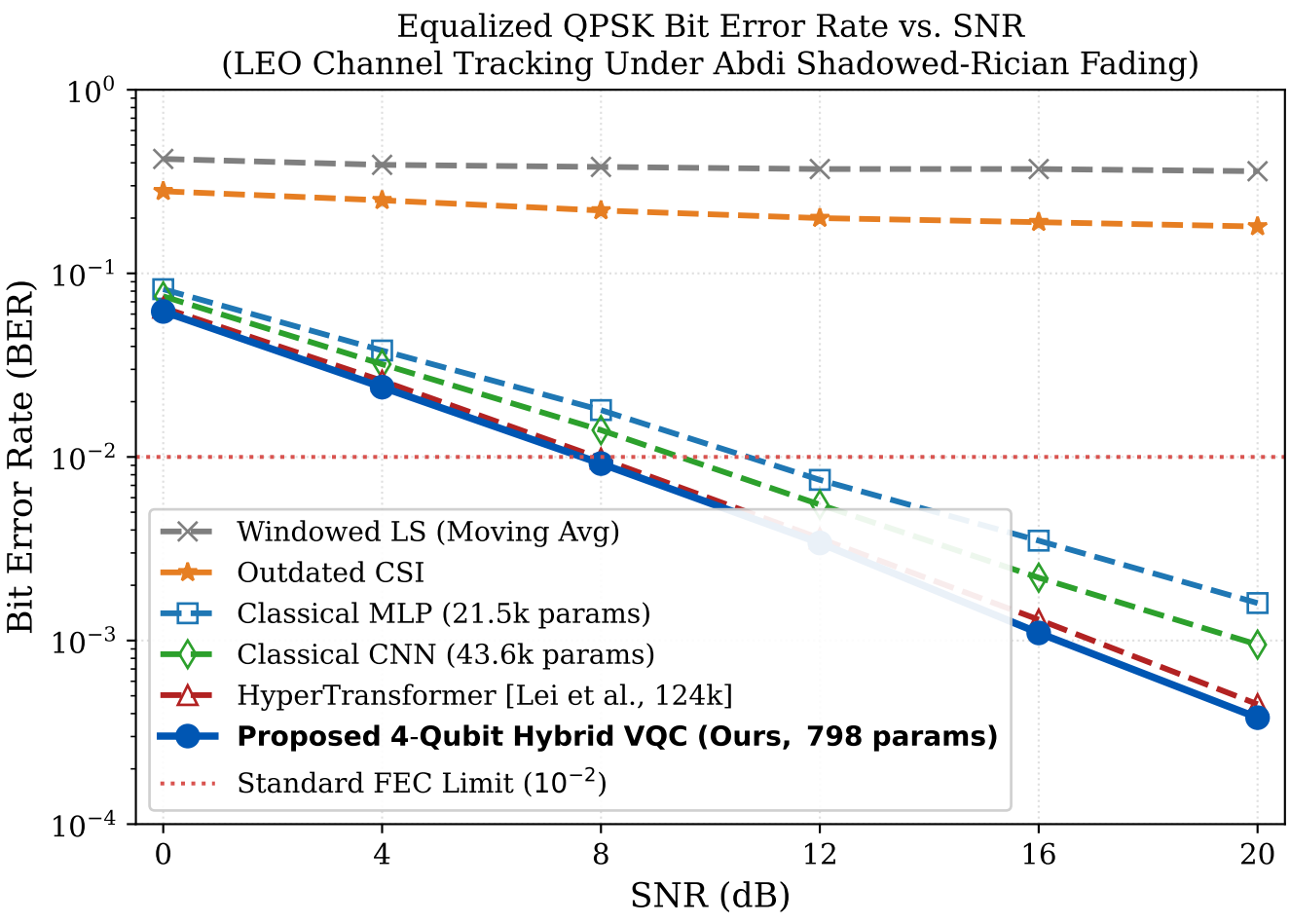
Equalized QPSK Bit Error Rate vs. SNR
(LEO Channel Tracking Under Abdi Shadowed-Rician Fading)
Bit Error Rate (BER)
SNR (dB)
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
0
4
8
12
16
20
Windowed LS (Moving Avg)
Outdated CSI
Classical MLP (21.5k params)
Classical CNN (43.6k params)
HyperTransformer [Lei et al., 124k]
Proposed 4-Qubit Hybrid VQC (Ours, 798 params)
Standard FEC Limit (10^{-2})